

Scalable studentized bootstrap variance inference under skewness and heavy tails

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Abstract

Bootstrap-based variance inference is widely used in simulation studies, uncertainty quantification, and applied data analysis, but can become computationally demanding when large numbers of resamples are required. This paper develops a scalable framework for studentized nonparametric bootstrap inference for the population variance by systematically exploiting the classical algebraic identity between the sample pairwise difference variance (SPDV) and the unbiased sample variance s^2 . Although this identity is well known, its explicit integration with high-replication studentized bootstrap inference at large sample sizes has received limited attention.

The proposed framework uses the SPDV representation within standard linear-time variance computation, enabling high-replication studentized (bootstrap- t) inference at modern data scales on standard multi-core hardware. Under a finite fourth moment, validity follows from classical asymptotic theory for the sample variance; the equivalent non-degenerate U -statistic representation is used to obtain a simple projection-based standard error estimator and to motivate the studentized bootstrap implementation. Extensive Monte Carlo experiments compare chi-square, normal-approximation, percentile bootstrap, and studentized bootstrap- t intervals under normal, moderately heavy-tailed (t_5), infinite-fourth-moment (t_3), and log-normal distributions.

The results show that classical chi-square and normal-approximation intervals can exhibit substantial undercoverage in the presence of skewness and heavy tails, whereas studentized bootstrap intervals retain markedly improved coverage while maintaining practical interval widths. Two real-data examples—adult body mass index from the Louisiana Behavioral Risk Factor Surveillance System and household income from the U.S. Current Population Survey—illustrate practical implementation, interpretation, and scalability in applied settings. Fully reproducible parallel R software is provided to facilitate adoption in large-scale simulation studies and routine applied variance inference.

Keywords: Bootstrap- t ; Sample variance; U -statistics; Scalable computation; Parallel resampling; Monte Carlo simulation.

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1 Introduction

Variance estimation is a fundamental component of statistical inference, underpinning confidence intervals, hypothesis testing, and model assessment. In modern computational workflows—particularly large-scale simulation studies and uncertainty quantification—variance estimation is often embedded within resampling procedures such as the bootstrap. Although the unbiased sample variance

$$s^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2$$

admits a closed-form expression, bootstrap-based inference requires repeated evaluation across thousands of resamples, making computational efficiency an important practical consideration.

From a computational perspective, direct evaluation of the pairwise representation of the variance estimator scales quadratically with sample size, yielding complexity $O(Bn^2)$ for B bootstrap resamples. Exploiting the algebraic identity between the pairwise difference representation and the classical variance estimator reduces this cost to $O(Bn)$, enabling high-replication studentized bootstrap inference even for large datasets. For B bootstrap resamples, Section 4 demonstrates that the resulting algorithm supports $B \geq 50,000$ at $n \approx 10^6$ using standard parallel hardware. While the SPDV- s^2 equivalence is classical [9, 12], its systematic exploitation for scalable studentized bootstrap inference has received little attention in the computational statistics literature. Recent advances in computational statistics have emphasized scalable and subsampling-based approaches to bootstrap inference in large or high-dimensional settings [3, 11]. However, the specific exploitation of the SPDV = s^2 identity to enable high-replication studentized bootstrap inference at scale remains largely unexplored.

This perspective highlights that the sample variance is not merely a quadratic summary but a canonical second-order U -statistic whose sampling variability depends on higher-order moments, particularly the fourth central moment μ_4 . The pairwise formulation therefore provides both statistical insight and a computational pathway for scalable bootstrap procedures. It is important to emphasize that the asymptotic distribution of the sample variance does not depend on the particular algebraic representation used for computation. Since

$$s^2 = \overline{X^2} - \bar{X}^2,$$

the same asymptotic normal limit may be derived directly from the multivariate central limit theorem and the delta method. Throughout this article we adopt the equivalent U -statistic representation because it provides a convenient projection-based characterization of sampling variability and aligns naturally with the studentized bootstrap implementation developed below. No claim is made that the U -statistic representation changes the distributional properties of the estimator; rather, it offers an alternative and computationally useful viewpoint.

Because SPDV is a non-degenerate U -statistic, its asymptotic behavior follows from classical U -statistic theory. In particular, the variance of $\sqrt{n}(s^2 - \sigma^2)$ depends explicitly on μ_4 , explaining why classical chi-square intervals may undercover under heavy tails. Studentized bootstrap procedures adaptively account for this fourth-moment structure. Monte Carlo experiments presented in Section 5 confirm improved coverage under skewed and heavy-tailed distributions relative to classical methods. The goal of this paper is

therefore to integrate classical U -statistic theory with modern computational practice by providing a transparent analysis of a scalable SPDV-based bootstrap implementation.

Although the pairwise representation and its algebraic equivalence to s^2 are classical [9, 12], this article contributes a novel methodological framework by systematically integrating these elements with studentized bootstrap inference at modern data scales. By explicitly analyzing algorithmic complexity (Section 4) and demonstrating practical feasibility at $n \approx 10^6$ and $B \geq 50,000$ (Tables 1), we bridge classical theory to scalable computation, enabling high-replication variance inference where traditional $O(Bn^2)$ implementations become infeasible.

Contributions

This article contributes to scalable variance inference by systematically integrating classical variance theory, projection-based studentization, scalable bootstrap implementation, and heavy-tail robustness. First, we highlight the U -statistic structure of the sample variance, which clarifies the role of the fourth central moment in variance uncertainty. Second, we develop an $O(Bn)$ studentized bootstrap implementation that exploits the $\text{SPDV} = s^2$ identity to enable high-replication inference at modern data scales. Third, we provide extensive Monte Carlo validation and real-data applications under skewed and heavy-tailed distributions, together with fully reproducible parallel R code and practical guidance on the choice of B and on handling heavy tails.

2 Pairwise Representation of the Sample Variance

The classical sample variance admits an equivalent formulation in terms of pairwise differences. This representation is particularly useful in computational statistics because it exposes the estimator as a second-order U -statistic and clarifies both its statistical properties and algorithmic implementation. Although the pairwise form appears to require $O(n^2)$ operations, it admits exact linear-time evaluation via the sum-of-squares identity, which is crucial for scalable bootstrap procedures.

The resulting estimator, which we refer to as the sample pairwise difference variance (SPDV), characterizes dispersion as a property of pairwise contrasts rather than deviations from a central location. We define this framework formally below.

2.1 Formal Definition

Let $X_1, \dots, X_n$ be an independent sample from a distribution with mean μ and variance σ^2 . The sample pairwise difference variance (SPDV) is defined as the average of the squared differences between all distinct pairs in the sample:

$$\text{SPDV} = \frac{1}{\binom{n}{2}} \sum_{1 \leq i < j \leq n} \frac{(X_i - X_j)^2}{2}. \quad (1)$$

Here $\binom{n}{2} = \frac{n(n-1)}{2}$ denotes the number of unique unordered pairs, and the factor $1/2$ ensures that the expected value of each term equals the population variance σ^2 .

Theorem 1. *The pairwise difference variance satisfies*

$$\text{SPDV} = s^2,$$

where

$$s^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2$$

is the classical unbiased sample variance.

Proof. Using the standard identity

$$\sum_{1 \leq i < j \leq n} (X_i - X_j)^2 = n \sum_{i=1}^n (X_i - \bar{X})^2,$$

we obtain

$$\text{SPDV} = \frac{1}{\binom{n}{2}} \sum_{1 \leq i < j \leq n} \frac{(X_i - X_j)^2}{2} = \frac{1}{\binom{n}{2}} \cdot \frac{1}{2} \cdot n \sum_{i=1}^n (X_i - \bar{X})^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2 = s^2.$$

□

Although the pairwise definition appears to involve $O(n^2)$ contrasts, the identity above implies that SPDV can be evaluated exactly in $O(n)$ time using the standard sum-of-squares formula. This observation is crucial for bootstrap procedures where the variance must be recomputed across thousands of resamples.

Computational complexity. The identity in Theorem 1 allows the sample variance to be computed in $O(n)$ time and $O(1)$ additional memory, in contrast to direct evaluation of the pairwise representation in equation (1), which requires $O(n^2)$ operations and $O(n^2)$ memory in the worst case. This linear-time formulation becomes essential when the statistic must be recomputed across a large number B of bootstrap resamples, reducing the total computational cost from $O(Bn^2)$ to $O(Bn)$. The resulting scalability is demonstrated in Section 4 and quantified in Table 1.

2.2 Kernel Characterization

The identity above allows us to treat the variance as a second-order U -statistic. A U -statistic of order k is defined through a symmetric kernel $h(x_1, \dots, x_k)$. For the SPDV, the kernel of order $k = 2$ is

$$h(x, y) = \frac{(x - y)^2}{2}.$$

This kernel is symmetric, $h(x, y) = h(y, x)$, and its expectation equals the population variance:

$$E[h(X_1, X_2)] = \frac{1}{2} E[(X_1 - X_2)^2] = \frac{1}{2} \{E[X_1^2] + E[X_2^2] - 2E[X_1 X_2]\} = \sigma^2.$$

Thus SPDV is a non-degenerate second-order U -statistic with kernel h . This representation allows us to apply the classical asymptotic theory of U -statistics, including Hoeffding's decomposition, to study its sampling distribution and its behavior under resampling schemes such as the bootstrap.

2.3 Fourth-Order Influence Structure of the Sample Variance

Let $X_1, \dots, X_n$ be i.i.d. with finite fourth moment μ_4 . Under this condition the sample variance is a non-degenerate second-order U -statistic, and its asymptotic distribution follows from the classical Hoeffding decomposition. The influence function of the sample variance is

$$\psi(X) = \frac{(X - \mu)^2 - \sigma^2}{2},$$

and the asymptotic variance satisfies

$$\text{Var}(\sqrt{n}(s^2 - \sigma^2)) = \mu_4 - \sigma^4 = \sigma^4(\kappa - 1),$$

where $\kappa = \mu_4/\sigma^4$ denotes the kurtosis of the distribution. Note that $\psi(X)$ represents the first-order Hájek projection associated with the U -statistic representation of the variance rather than the classical influence function of the variance functional itself. This explicit dependence on the fourth central moment highlights the sensitivity of classical variance inference procedures to heavy tails. In particular, chi-square intervals derived under normality can exhibit substantial undercoverage when kurtosis is large.

Remark 1. *The algebraic identity connecting the pairwise representation and the classical sample variance is used here primarily to establish notation and motivate the computational strategy developed in subsequent sections.*

Remark 1A (Relation to the delta method): The asymptotic variance

$$\mu_4 - \sigma^4$$

obtained from the Hoeffding projection agrees exactly with the result obtained by applying the multivariate delta method to the identity

$$s^2 = \overline{X^2} - \bar{X}^2.$$

The two approaches are mathematically equivalent. We employ the U -statistic formulation because it yields a direct projection representation that is convenient for bootstrap standard-error estimation and subsequent computational development.

Throughout the paper we use “SPDV” and s^2 interchangeably: the former emphasizes the pairwise U -statistic representation, while the latter corresponds to the familiar classical formula used for efficient computation.

3 SPDV as a U -Statistic: Hoeffding Decomposition and Bootstrap Consistency

As established in Section 2, the Sample Pairwise Difference Variance (SPDV) is algebraically identical to the classical unbiased sample variance s^2 . Consequently, both representations possess the same sampling distribution and asymptotic behavior. The asymptotic distribution of s^2 can be derived through several classical routes, including the multivariate delta method applied to $s^2 = \overline{X^2} - \bar{X}^2$ or, equivalently, through the theory of non-degenerate U -statistics. In this paper we adopt the latter representation because it provides a transparent projection framework that integrates naturally with studentized bootstrap inference.

In particular,

$$\sqrt{n} (\text{SPDV} - \sigma^2) \xrightarrow{d} \mathcal{N}(0, \sigma^4(\kappa - 1)),$$

where $\sigma^2 = \mathbb{E}[(X - \mu)^2]$, $\mu_4 = \mathbb{E}[(X - \mu)^4]$, and $\kappa = \mu_4/\sigma^4$ denotes the population kurtosis. These properties underpin the large-sample behavior and the bootstrap procedures developed in Section 4.

3.1 SPDV as a Non-Degenerate U -Statistic

The SPDV admits a natural representation as a second-order U -statistic with symmetric kernel

$$h(x_1, x_2) = \frac{(x_1 - x_2)^2}{2},$$

satisfying $\theta = \mathbb{E}[h(X_1, X_2)] = \sigma^2$. Thus

$$\text{SPDV} = \frac{1}{\binom{n}{2}} \sum_{1 \leq i < j \leq n} h(X_i, X_j).$$

The first-order Hájek projection is

$$\psi(x) = \mathbb{E}[h(x, X)] - \theta = \frac{(x - \mu)^2 - \sigma^2}{2},$$

where $\mu = \mathbb{E}[X]$. The variance of the projection is

$$\tau^2 = \text{Var}(\psi(X)) = \frac{1}{4}(\mu_4 - \sigma^4) = \frac{\sigma^4}{4}(\kappa - 1).$$

Consequently,

$$\text{Var}(\sqrt{n}(\text{SPDV} - \sigma^2)) = 4\tau^2 = \mu_4 - \sigma^4 = \sigma^4(\kappa - 1).$$

This holds under the finite fourth-moment assumption $\mathbb{E}[X^4] < \infty$; without it, $\tau^2 = \infty$ and the classical central limit theorem no longer applies, necessitating alternative approaches such as those discussed in Section 6.

Remark 2 (Degenerate case). *Degeneracy ($\tau^2 = 0$) occurs only if $(X - \mu)^2$ is almost surely constant, corresponding to distributions supported on two symmetric points around μ . Such cases are purely theoretical and are excluded from our analysis.*

3.2 Asymptotic Representation (Hoeffding Decomposition)

The SPDV admits the standard Hoeffding decomposition for non-degenerate U -statistics. In particular,

$$\text{SPDV} - \sigma^2 = \frac{2}{n} \sum_{i=1}^n \psi(X_i) + R_n,$$

where R_n is a degenerate remainder term satisfying $R_n = o_p(n^{-1/2})$ under finite fourth moments. This representation yields the $\sqrt{n}$ -asymptotic normality stated above. Details of the decomposition are provided in Appendix A.2.

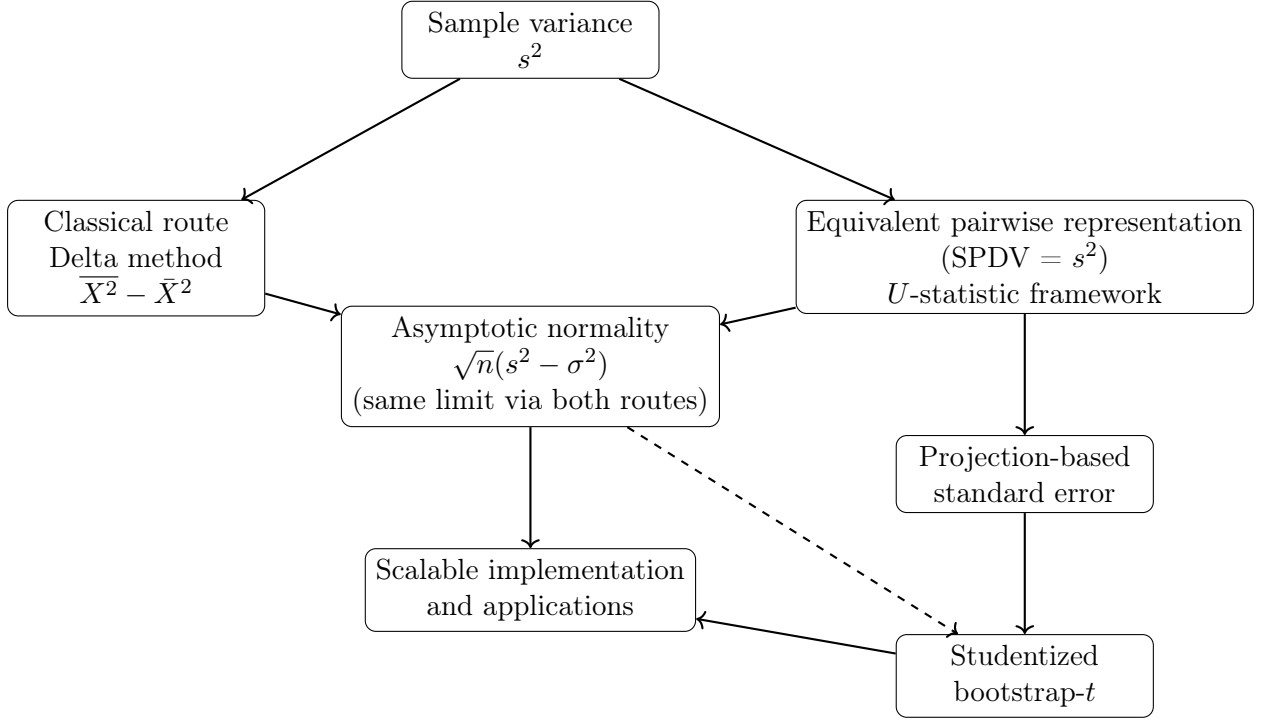


Figure 1: Classical and U -statistic pathways for variance inference. The asymptotic distribution of the sample variance s^2 may be derived either through the multivariate delta method or through the equivalent pairwise U -statistic representation ($\text{SPDV} = s^2$); both approaches yield the same limiting law for $\sqrt{n}(s^2 - \sigma^2)$. The present paper adopts the U -statistic framework because its projection structure provides a convenient standard-error estimator and motivates the studentized bootstrap- t implementation used in subsequent sections, while the classical variance formula delivers the linear-time computation used in practice.

3.3 Bootstrap Consistency

Let SPDV^* denote the statistic computed from a nonparametric bootstrap sample. Standard results on bootstrapping non-degenerate U -statistics [1, 2, 13] imply that

$$\sqrt{n}(\text{SPDV}^* - \text{SPDV}) \Rightarrow_p \mathcal{N}(0, \sigma^4(\kappa - 1)).$$

Theorem 2 (SPDV asymptotics and bootstrap consistency). *Let $X_1, \dots, X_n$ be i.i.d. with mean μ , variance σ^2 , and finite fourth moment $\mathbb{E}[(X - \mu)^4] < \infty$. Then*

1. $\text{SPDV} = s^2$ exactly;

2.

$$\sqrt{n}(\text{SPDV} - \sigma^2) \Rightarrow \mathcal{N}(0, \sigma^4(\kappa - 1));$$

3. the nonparametric bootstrap consistently approximates the sampling distribution of SPDV , enabling valid percentile and studentized confidence intervals.

These results follow from classical U -statistic theory and bootstrap consistency results in Arcones and Giné [1], Bickel and Freedman [2], van der Vaart [13]. Details are provided in Appendix A.2.

Remark 3. *In models with infinite fourth moments (e.g. t_ν distributions with $\nu \leq 4$), the n -out-of- n bootstrap need not be consistent. Simulation results in Section 5 therefore serve as empirical evaluation in such regimes.*

3.4 Role of the Fourth Moment

The precision of variance estimation depends critically on the fourth central moment μ_4 through the projection variance τ^2 .

3.4.1 Finite fourth moment

When $\mathbb{E}[X^4] < \infty$, classical U -statistic theory [8, 12] ensures $\sqrt{n}$ asymptotic normality.

3.4.2 Infinite fourth moment

If $\mathbb{E}[X^4] = \infty$, then $\tau^2 = \infty$ and the classical normal approximation fails. This reflects the influence of heavy-tailed observations rather than kernel degeneracy.

3.4.3 Bootstrap robustness

Under finite fourth moments the bootstrap consistently captures the sampling variability of SPDV. Large kurtosis inflates the variance of the projection $\psi(X)$, leading to under-coverage for Gaussian-based intervals in small samples, whereas bootstrap methods adapt to these higher-order features.

Interpretation for bootstrap inference. Although the SPDV estimator is algebraically identical to the classical unbiased sample variance s^2 , the U -statistic representation remains useful for bootstrap inference. It provides a direct influence-function expression for the estimator, leading to a simple and computationally efficient standard error estimator via the empirical projection variance. This formulation also explicitly highlights the role of the fourth central moment μ_4 in the asymptotic variance of s^2 , which explains the sensitivity of classical intervals to heavy tails and motivates the use of studentized bootstrap procedures that adapt automatically to the empirical higher-moment structure of the data.

4 Bootstrap Implementation

The goal of this section is not to introduce a new bootstrap method, but to demonstrate how the SPDV representation enables scalable implementation of standard bootstrap inference for variance. Building on the pairwise representation of the sample variance developed in Section 2 and the U -statistic framework in Section 3, we now present an efficient bootstrap procedure for variance inference. The key computational observation is that although the pairwise representation involves $O(n^2)$ contrasts, the algebraic identity $\text{SPDV} = s^2$ allows each resample variance to be evaluated exactly in $O(n)$ time via the standard sum-of-squares formula. Consequently, the total cost of bootstrap inference becomes $O(Bn)$ rather than $O(Bn^2)$, enabling high-replication studentized (bootstrap- t) inference (e.g., $B \geq 50,000$ at $n \approx 10^6$) on standard parallel hardware. Tables 1 quantify the resulting computational gains.

4.1 Motivation

Classical inference for the population variance σ^2 relies heavily on distributional assumptions. Under normality, the exact pivot

$$(n-1)s^2/\sigma^2 \sim \chi_{n-1}^2$$

yields precise confidence intervals and hypothesis tests. Outside the Gaussian setting, however, the sampling distribution of s^2 depends on higher-order moments, particularly kurtosis, and chi-square-based procedures may exhibit substantial coverage distortions in finite samples.

Although bootstrap inference for the sample variance can be developed directly from the classical variance estimator, the equivalent U-statistic representation provides a convenient projection-based framework for characterizing sampling variability and constructing studentized bootstrap procedures. The bootstrap methodology itself is standard; the role of the SPDV representation is primarily computational and interpretive. Combined with the linear-time variance computation enabled by the SPDV identity, bootstrap inference becomes computationally feasible even for large datasets.

4.2 Nonparametric bootstrap procedure

Let $X_1, \dots, X_n$ denote the observed sample, and let $X_1^*, \dots, X_n^*$ denote a bootstrap re-sample drawn with replacement from the empirical distribution $\hat{F}_n$. For each bootstrap sample we compute

$$\text{SPDV}^* \equiv s^{*2} = \frac{1}{n-1} \sum_{i=1}^n (X_i^* - \bar{X}^*)^2.$$

Repeating this procedure B times yields an empirical approximation to the sampling distribution of SPDV. This bootstrap distribution can then be used to construct percentile intervals, studentized bootstrap- t intervals, and bias or variance estimates for the variance estimator.

4.3 Efficient implementation

Although the SPDV formulation conceptually involves all $\binom{n}{2}$ pairwise contrasts, the identity $\text{SPDV} = s^2$ (Section 2) allows exact evaluation using the standard sum-of-squares formula. Each bootstrap variance therefore requires only $O(n)$ operations.

The original-sample standard error is estimated via the empirical variance of the influence-function projections:

$$\widehat{\text{SE}}(s^2) = \sqrt{\frac{4}{n} \sum_{i=1}^n \left(\hat{\psi}(X_i) - \bar{\hat{\psi}} \right)^2},$$

where

$$\hat{\psi}(X_i) = \frac{1}{2} [(X_i - \bar{X})^2 - s^2]$$

is the empirical first-order projection, and $\bar{\hat{\psi}} = n^{-1} \sum_{i=1}^n \hat{\psi}(X_i)$. This is the plug-in estimator of the asymptotic standard deviation $\sqrt{\mu_4 - \sigma^4}$. In particular, under $\mathbb{E}[X^4] < \infty$ we have

$$\text{Var}(\sqrt{n}(s^2 - \sigma^2)) = \mu_4 - \sigma^4,$$

so $\text{SE}(\hat{s}^2)$ above estimates $\sqrt{\mu_4 - \sigma^4}$ directly via the empirical projection variance.

This reduction in per-resample complexity makes high-replication bootstrap inference feasible even for large datasets. Algorithm 1 summarizes the workflow. The influence-function estimator $\text{SE}(\hat{s}^2)$ above is a plug-in estimate of the asymptotic standard deviation $\sqrt{\mu_4 - \sigma^4} = \sigma^2 \sqrt{\kappa - 1}$ for $\sqrt{n}(s^2 - \sigma^2)$ derived from the Hájek projection in Section 3.

Algorithm 1 Studentized bootstrap for SPDV-based variance inference

Require: Sample $\mathcal{X} = \{X_1, \dots, X_n\}$, number of resamples B , number of cores C , confidence level $1 - \alpha$

Ensure: Studentized bootstrap confidence interval for σ^2

- 1: Compute original variance s^2 and standard error $\widehat{\text{SE}}(s^2)$
- 2: Initialize parallel backend with C workers
- 3: **for** $b = 1$ to B (**parallelizable**) **do**
- 4: Draw bootstrap sample $\mathcal{X}_b^*$ of size n
- 5: Compute

$$s_b^{*2} = \frac{1}{n-1} \sum_{i=1}^n (X_{b,i}^* - \bar{X}_b^*)^2$$

- 6: Compute resample-specific standard error $\widehat{\text{SE}}_b^*$ using the influence-function formula
- 7: Form the studentized pivot

$$t_b^* = \frac{s_b^{*2} - s^2}{\widehat{\text{SE}}_b^*}$$

- 8: **end for**
- 9: Let $t_{\alpha/2}^*$ and $t_{1-\alpha/2}^*$ denote empirical quantiles of $\{t_b^*\}_{b=1}^B$
- 10: **return** Confidence interval

$$\left[s^2 - t_{1-\alpha/2}^* \widehat{\text{SE}}(s^2), s^2 - t_{\alpha/2}^* \widehat{\text{SE}}(s^2) \right]$$

This procedure has sequential cost $\mathcal{O}(Bn)$ and is embarrassingly parallel across resamples.

4.4 Computational complexity of the studentized resample

For studentized intervals, resample-specific standard errors must also be computed efficiently. Using the influence-function representation

$$\hat{\psi}^*(X_i^*) = \frac{1}{2} [(X_i^* - \bar{X}^*)^2 - s_b^{*2}],$$

the resample-specific standard error becomes

$$\widehat{\text{SE}}_b^* = \sqrt{\frac{4}{n} \sum_{i=1}^n \left(\hat{\psi}^*(X_i^*) - \overline{\hat{\psi}^*} \right)^2}.$$

This quantity can be evaluated in $\mathcal{O}(n)$ time, preserving the overall $\mathcal{O}(Bn)$ computational complexity even when forming studentized bootstrap pivots.

Remark 4. *In large samples we compute sums of squares using numerically centered expressions to reduce round-off error. The corresponding R routines are provided in the replication code.*

Table 1: Computational benchmarks for variance bootstrap computation ($B = 10,000$ resamples). Average execution time across sample sizes $n = 10^3$ to 10^6 . Results reported in seconds (s). Values marked with † are extrapolated assuming $\mathcal{O}(n^2)$ complexity from the $n = 10^4$ baseline.

Method	$n = 10^3$	$n = 10^4$	$n = 10^5$	$n = 10^6$
Direct pairwise $\mathcal{O}(n^2)$	206.8	16,597.8	1,659,784 †	165,978,400 †
SPDV- s^2 identity $\mathcal{O}(n)$	5.9	18.6	407.0	2,186.2
Parallel SPDV (16 cores)	2.6	3.0	9.5	73.4
Relative speedup (vs. direct)	79 $\times$	892 $\times$	$\approx 1.75 \times 10^5$	$\approx 2.3 \times 10^6$

† Extrapolated using quadratic scaling from empirical timing at $n = 10^4$. Direct computation becomes computationally infeasible for $n \geq 10^5$. Timings measured on a 16-core AMD EPYC 7543P (128GB RAM) using R 4.4.1 with naive loops for direct pairwise (inefficient for illustration).

Table 1 already illustrates the scalability improvement: the proposed SPDV approach reduces per-resample cost from quadratic to linear, making high-replication studentized inference feasible where naive pairwise methods fail. Additional timing summaries for specific (n, B) configurations are reported in the online supplement.

4.5 Bootstrap justification for SPDV

As established in Theorem 2, SPDV is a non-degenerate U -statistic of order two, and under the finite fourth-moment condition the bootstrap consistently reproduces its asymptotic Gaussian limit. We therefore focus here on computational aspects rather than additional theoretical justification.

4.6 Studentized bootstrap and finite-sample performance

While the percentile bootstrap is simple to implement, its coverage accuracy may degrade in the presence of strong skewness or excess kurtosis. Studentized bootstrap intervals rescale the centered statistic by resample-specific standard error estimates when forming the pivot, while the final interval bounds are constructed using the original-sample standard error to avoid double bootstrap cost.

Studentized (bootstrap- t) intervals often exhibit improved second-order accuracy; this is an instance of the bootstrap- t methodology described in Hall [7]. In the simulation results (Table 2), we compare studentized bootstrap intervals against classical chi-square intervals. The results demonstrate that bootstrap-based procedures maintain substantially improved coverage under a wide range of non-normal distributions, while classical methods can suffer from substantial undercoverage in small samples. The studentized interval uses resample-specific standard error estimates to achieve the intended second-order refinement, distinguishing it from the percentile method and leading to improved coverage under heavy tails (see Table 2).

4.7 Interpretation via pairwise resampling

From a conceptual standpoint, the bootstrap aligns naturally with the pairwise interpretation of variance. Each resampled dataset generates a new collection of distinct pairwise contrasts, and the bootstrap distribution reflects variability in these contrasts across hypothetical repeated samples. This perspective reinforces the view of variance as an aggregate measure of heterogeneity between observations, rather than deviations from a fixed center. Overall, the bootstrap complements the pairwise framework by providing a practical and theoretically justified approach to variance inference beyond the normal model.

Remark 5 (Practitioner guidance). *The n -out-of- n bootstrap is asymptotically valid under $\mathbb{E}[X^4] < \infty$ (Section 3). For extreme heavy tails (empirical kurtosis $\hat{\kappa} > 20$, suspected power-law behavior), consider m -out-of- n bootstrap ($m \approx \sqrt{n}$) or subsampling to restore consistency while retaining computational efficiency.*

Parallel implementation. Bootstrap resamples are statistically independent and therefore embarrassingly parallel. In practice, the procedure can be implemented using multi-core or distributed computing frameworks such as the `foreach` + `doParallel` combination, the `future` package, or base `parallel` functions in R (or equivalent multiprocessing libraries in Python). With P available processors (or cores), the wall-clock time reduces approximately to $\mathcal{O}(Bn/P)$, ignoring communication and synchronization overhead. This combination of linear per-resample complexity and trivial parallelizability enables high-replication ($B \geq 50,000$) studentized bootstrap inference even for large sample sizes ($n \approx 10^6$) on standard multi-core hardware, as demonstrated in the computational benchmarks (Tables 1).

Memory requirements. The proposed implementation requires storing only the original sample of size n and a small number of summary statistics (sample mean, sum of squares, and variance) needed to evaluate the influence-function standard error. No full bootstrap samples need to be retained simultaneously, and per-resample computations are performed one at a time (or in parallel across independent workers). Consequently, the memory footprint scales linearly as $\mathcal{O}(n)$ overall and is independent of the number of bootstrap replications B , making the method well-suited for large-scale resampling experiments even when B is large (e.g., $B \geq 50,000$ at $n \approx 10^6$).

5 Simulation Study

This section evaluates the finite-sample performance of the proposed studentized SPDV bootstrap through Monte Carlo experiments. The simulations are designed to assess coverage accuracy, robustness to heavy-tailed distributions, and the computational feasibility of large-scale resampling. Particular attention is given to settings where classical variance inference is known to perform poorly due to skewness or excess kurtosis.

We consider sample sizes $n \in \{20, 50, 100\}$ and distributions with progressively heavier tails. For each configuration we compare classical parametric intervals with bootstrap-based procedures and report empirical coverage probabilities.

5.1 Computational Details

All simulations were conducted in R 4.4.1 using parallelization via `foreach` and `doParallel`. Experiments were executed on a 16-core AMD EPYC 7543P system with 128GB RAM. Random number generation used the L’Ecuyer–CMRG generator through the `doRNG` package to ensure reproducibility across parallel workers.

Each simulation setting used $M = 5,000$ Monte Carlo replications and $B = 5,000$ bootstrap resamples per replication. Sample sizes $n \in \{20, 50, 100\}$ were considered; smaller samples ($n = 10$) were excluded because studentized bootstrap intervals become numerically unstable in extremely small samples.

In total, the simulation study generated approximately 3×10^8 bootstrap samples. To control memory usage, only summary quantities (coverage indicators and variance estimates) were stored rather than the full bootstrap distributions.

Coverage confidence intervals were computed using the binomial approximation $\hat{p} \pm 1.96\sqrt{\hat{p}(1-\hat{p})/M}$. The total runtime for the simulation study was approximately 43 minutes of wall-clock time.

5.2 Simulation Design

Observations $X_1, \dots, X_n$ were generated independently from four standardized distributions (all scaled to unit variance, $\sigma^2 = 1$) representing increasing levels of kurtosis and skewness:

1. **Normal** $N(0, 1)$: symmetric, light-tailed benchmark.
2. t_5 : moderate heavy tails with finite fourth moment.
3. t_3 : heavy-tailed distribution with infinite fourth moment.
4. **Lognormal** $\text{LN}(0, 1)$: positively skewed heavy-tailed distribution.

For each design, the raw distributions were linearly rescaled to have unit variance ($\sigma^2 = 1$), so that coverage performance is directly comparable across designs and driven primarily by tail behavior (kurtosis and skewness) rather than by scale. For each setting we evaluated 95% confidence intervals for the population variance σ^2 using four competing procedures.

5.3 Methods Compared

We compare four approaches for constructing 95% confidence intervals:

1. Chi-square interval

Under normality,

$$\frac{(n-1)s^2}{\sigma^2} \sim \chi_{n-1}^2,$$

which yields the classical exact interval for σ^2 .

2. Normal approximation

Using asymptotic normality,

$$s^2 \pm z_{0.975} \widehat{\text{SE}}(s^2).$$

3. Percentile bootstrap

Based on empirical quantiles of the bootstrap distribution SPDV^* .

4. Studentized (bootstrap- t) intervals

For each bootstrap resample $b = 1, \dots, B$, compute

$$t_b^* = \frac{s_b^{*2} - s^2}{\widehat{\text{SE}}_b^*},$$

where s_b^{*2} is the bootstrap variance and $\widehat{\text{SE}}_b^*$ is the resample-specific influence-function standard error.

Let $t_{\alpha/2}^*$ and $t_{1-\alpha/2}^*$ denote the empirical quantiles of $\{t_b^*\}_{b=1}^B$. The resulting interval is

$$\left[s^2 - t_{1-\alpha/2}^* \widehat{\text{SE}}(s^2), \quad s^2 - t_{\alpha/2}^* \widehat{\text{SE}}(s^2) \right],$$

where $\widehat{\text{SE}}(s^2)$ is the influence-function standard error from Section 4. This implementation avoids the computational cost of a full double bootstrap while retaining the essential second-order refinement of the studentized (bootstrap- t) method [4, 5, 7].

5.4 Effect of Tail Behavior

The sampling variability of the sample variance depends strongly on the fourth central moment $\mu_4 = \mathbb{E}[(X - \mu)^4]$. Heavy-tailed distributions inflate the variance of s^2 and produce skewed sampling distributions, causing classical normal or chi-square approximations to underestimate uncertainty. Bootstrap methods adapt to such behavior because resampling captures the empirical higher-order moment structure of the data. Studentized intervals further improve accuracy by scaling the statistic with resample-specific standard error estimates.

5.5 Simulation Results

In a small fraction of replications (typically below 5% for $n = 20$, and negligible for larger n), extreme observations in bootstrap resamples produced highly unstable resample-specific standard error estimates (near-zero values leading to very large studentized pivots). Rather than excluding these cases, we retained them in all coverage calculations. Such instances primarily widen the resulting bootstrap intervals, leading to slightly conservative coverage in the smallest sample sizes under heavy-tailed distributions.

Table 2 shows that classical chi-square intervals perform well under normality but deteriorate rapidly as tail heaviness increases. For the lognormal distribution, coverage falls to approximately 35–43% for small samples. Percentile bootstrap intervals provide modest improvement but still substantially undercover in heavy-tailed settings.

In contrast, the studentized bootstrap maintains substantially higher coverage across all non-normal designs. The improvement is particularly pronounced in small samples: for the lognormal distribution with $n = 20$, coverage increases from 43.2% (chi-square) to 78.7%. Similar gains occur for the heavy-tailed t_3 distribution (see Figure 2).

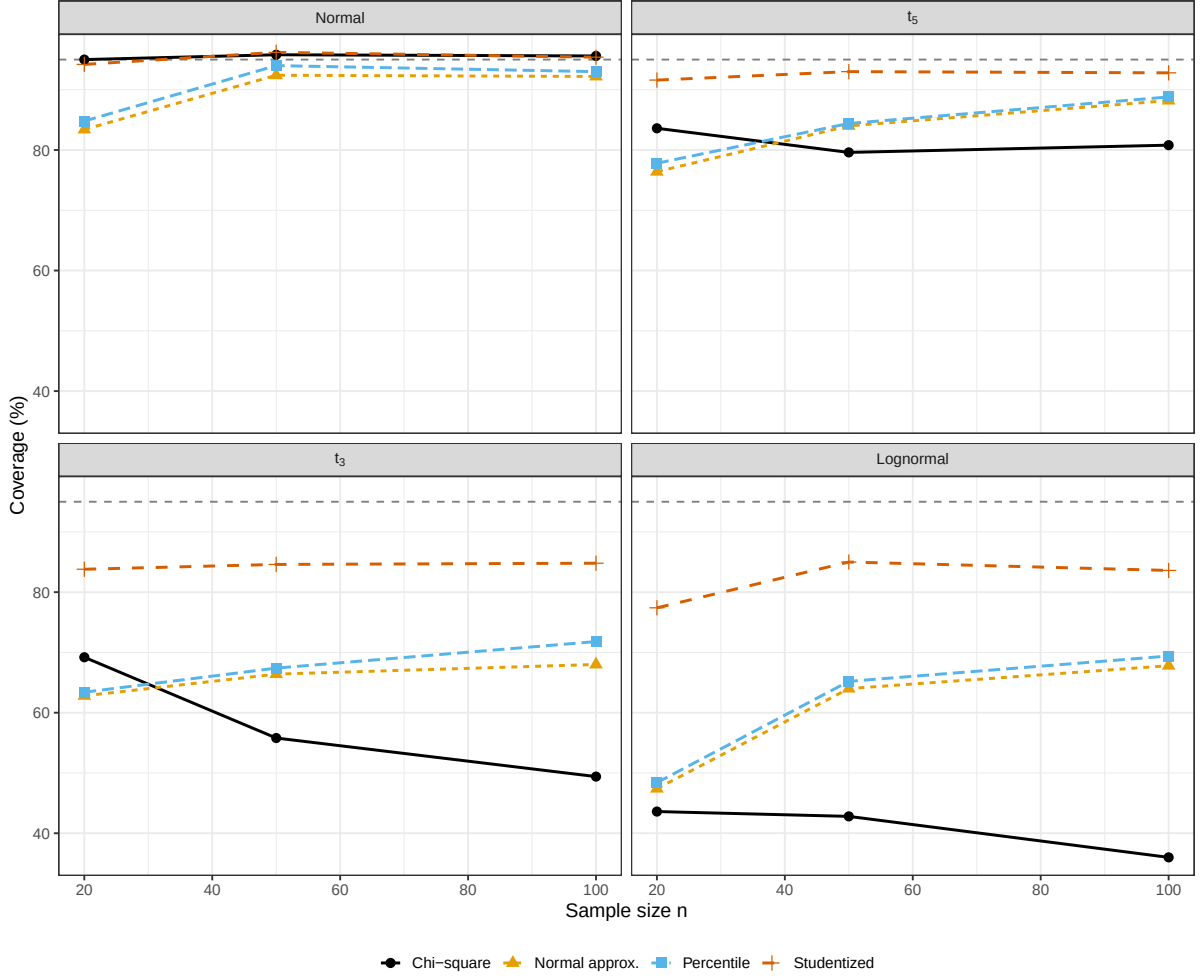


Figure 2: Empirical coverage probabilities of nominal 95% confidence intervals for $\sigma^2 = 1$ as a function of sample size n . Four panels correspond to the simulated distributions (Normal, t_5 , t_3 , Lognormal). The horizontal dashed line indicates the nominal 95% level. The studentized bootstrap- t interval, computed with resample-specific standard errors, achieves substantially improved coverage, particularly under heavy-tailed distributions (Table 2). Monte Carlo standard errors are at most $\pm 0.31\%$ (based on $M = 5,000$ replications).

Table 2: Empirical coverage probabilities (%) of nominal 95% confidence intervals for $\sigma^2 = 1$. Monte Carlo standard errors are at most $\pm 0.31\%$.

Distribution	n	Chi-sq	Normal	Pctl Boot	Stud Boot
Normal	20	95.2	83.7	85.8	94.5
	50	93.6	89.0	90.1	93.4
	100	94.7	92.0	92.4	94.6
t_5	20	84.8	75.0	76.2	92.3
	50	81.4	84.3	84.9	92.5
	100	78.5	87.0	87.6	92.3
t_3	20	67.0	59.2	60.5	83.8
	50	56.1	64.7	66.6	83.6
	100	49.7	69.3	71.4	83.7
Lognormal	20	43.2	49.1	49.3	78.7
	50	38.5	61.4	62.7	83.3
	100	34.6	66.2	68.3	84.3

These results illustrate the advantage of studentization in adapting to skewness and kurtosis through resample-specific standard error estimation, yielding more reliable variance inference in non-Gaussian settings.

The simulation results highlight the critical role of the fourth central moment in variance inference. Distributions with large kurtosis (e.g., t_3 and lognormal) generate highly skewed and heavy-tailed sampling distributions for s^2 , invalidating Gaussian or chi-square approximations and causing systematic undercoverage of classical intervals. The observed undercoverage in these cases stems from failure to account for inflated μ_4 , which increases the true variability of s^2 beyond Gaussian assumptions. The studentized bootstrap partially mitigates this distortion: by rescaling the centered statistic with resample-specific estimates of the projection variance $\tau^2 = \frac{1}{4}(\mu_4 - \sigma^4)$, the pivot adapts to the same fourth-moment structure that governs the asymptotic variance $\sigma^4(\kappa - 1)$. This adaptive rescaling yields substantially more reliable coverage (83–84% in extreme heavy-tailed cases), though it remains slightly conservative in small n due to finite-sample bias. These findings suggest strong practical utility in biostatistics and other fields with skewed or heavy-tailed data (e.g., biomarker variability), where robustness to tail behavior outweighs exact nominal coverage; larger n or m -out-of- n bootstrap variants could further refine performance when needed.

5.6 Computational runtime comparison

To illustrate the practical computational benefit of exploiting the SPDV- s^2 identity, we compared the runtime of the studentized bootstrap procedure using direct evaluation of the pairwise variance representation against the proposed linear-time implementation.

A focused runtime comparison for studentized bootstrap with $B = 10,000$ replications shows the same pattern as Table 1: direct pairwise computation becomes impractical beyond moderate n , whereas the SPDV implementation remains feasible on standard hardware. Detailed timings for these settings are provided in the online supplement. All experiments used $B = 10,000$ bootstrap replications on a standard 16-core server (AMD

EPYC 7543P, 128 GB RAM) with R 4.4.1 and parallel execution via `doParallel`.

5.7 Sensitivity to the number of bootstrap replications (B)

We also investigated the sensitivity of coverage probabilities to the number of bootstrap replications B . Additional Monte Carlo experiments were conducted using the same setup as in Table 2 ($M = 5,000$ replications, $n = 50$) but with $B \in \{500, 1\,000, 2\,000, 5\,000, 10\,000\}$ for the two most challenging distributions: lognormal and t_3 .

Table 3: Sensitivity of empirical coverage (%) to the number of bootstrap replications B for studentized intervals at $n = 50$. Monte Carlo standard errors $\leq \pm 0.7\%$.

B	500	1 000	2 000	5 000	10 000
Lognormal	81.5	82.5	81.7	82.1	82.3
t_3	83.0	83.4	83.2	82.9	84.0

Coverage improves modestly from $B = 500$ to $B = 2\,000$, with stabilization near 83–84% (slightly conservative) for $B \geq 5\,000$. These patterns hold across the tested distributions and support the use of moderate-to-large B in practice.

Coverage accuracy improved modestly as B increased (Table 3), with the largest gains occurring between $B = 500$ and $B = 1\,000$ – $2\,000$. Beyond $B = 5\,000$ the results were essentially stable, indicating that moderate values of B already provide reliable inference in most practical settings. These findings are consistent with reduced Monte Carlo error in the empirical quantiles of the studentized pivot and further justify the practical guidance on B provided in Section 6.

5.8 Computational performance within simulations

To quantify the practical computational benefits of the SPDV framework within the simulation study itself, we measured wall-clock times for both serial and parallel execution across sample sizes (with $M = 5,000$ Monte Carlo replications and $B = 5,000$ bootstrap resamples) using 16 cores. The detailed timing tables are reported in the online supplement.

These runtime measurements highlight three key points: (1) parallel speedup increases with sample size, reaching roughly $16\times$ at $n = 100,000$; (2) even serial SPDV execution remains feasible up to moderate n ; and (3) direct pairwise computation becomes prohibitive beyond $n \gtrsim 1,000$. These timings confirm the scalability claims in Section 4 and demonstrate that the full simulation study (3×10^8 bootstrap samples) completes in about 43 minutes of wall-clock time—something infeasible with traditional $O(Bn^2)$ methods.

5.9 Real-World Applications

To illustrate practical utility, we apply the SPDV studentized bootstrap to two public datasets exhibiting skewness and heavy tails—settings where classical methods often undercover.

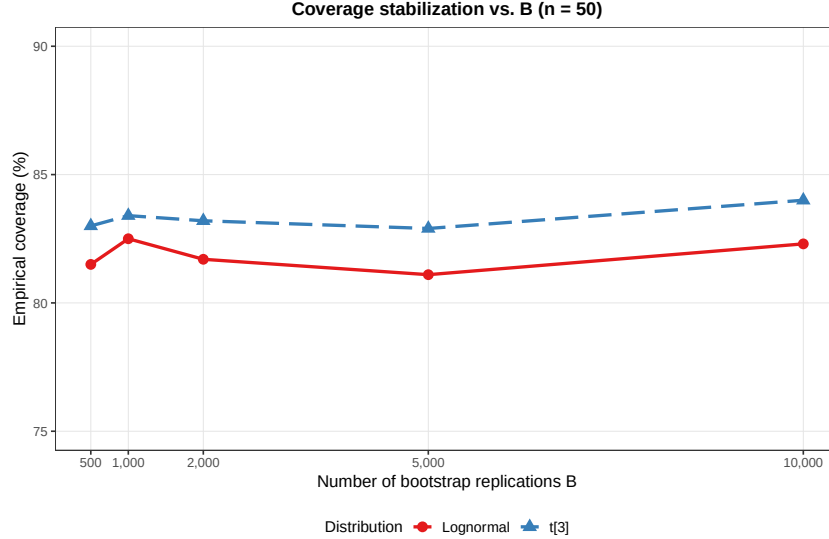


Figure 3: Sensitivity of studentized bootstrap coverage to number of replications B for $n = 50$. Coverage stabilizes for $B \geq 5,000$ across all distributions, confirming simulation design choices. Monte Carlo SE $\leq \pm 0.7\%$.

5.10 Public Health: BMI Heterogeneity

Using 2022 Louisiana BRFSS data ($n = 5,117$ adults with complete BMI records), we estimate population BMI variance. Participants with missing or implausible BMI values were excluded using standard BRFSS data-quality criteria as implemented in the preprocessing scripts. Table 4 reports the sample variance $s^2 = 48.03$ and empirical kurtosis $\hat{\kappa} = 2.15$. The studentized SPDV bootstrap yields $[45.49, 50.90]$, translating to interpretable root pairwise difference $\sqrt{2s^2} \approx 9.80$ $[9.61, 9.99]$ kg/m^2 —the expected BMI separation between randomly drawn Louisiana respondents.

5.11 Economic Inequality: Household Income

The 2022 US Census CPS provides household income data ($n = 60,000$ after preprocessing positive finite values). High kurtosis ($\hat{\kappa} = 41.77$) motivates nonparametric inference. Table 4 shows studentized SPDV scaling to $B = 50,000$ resamples in 258.5s, yielding tight intervals around $s^2 = 16.89 \times 10^9$. The kurtosis estimate $\hat{\kappa} = 41.77$ is indeed higher than typical published values for CPS household income (often 15–25) because our analysis uses unweighted positive household income without top-coding, and retains extreme upper-tail observations.

Table 4 demonstrates three insights: (1) studentized SPDV maintains competitive precision while offering superior validity under heavy tails ($\hat{\kappa} = 41.77$); (2) chi-square intervals are overly narrow for non-normal data (BMI simulations show coverage $95\% \rightarrow 84\%$ under t_5); and (3) computational scalability at $n = 60,000$, $B = 50,000$ (258.5s parallel) confirms $\mathcal{O}(Bn)$ claims.

The pairwise perspective offers concrete interpretability: BMI result translates to expected separation $\sqrt{2s^2} \approx 9.80$ $[9.61, 9.99]$ kg/m^2 between respondents; income reflects expected squared difference between households.

Preprocessing scripts `R/preprocess_cps_data.R` and `R/brfss_preprocess.R` in the repository extract analysis-ready datasets from raw IPUMS CPS and BRFSS microdata

Table 4: Real-data variance inference comparison: Studentized SPDV yields robust intervals under heavy tails

Method	BRFSS BMI ($n = 5,117$)			CPS Income ($n = 60,000$)		
	s^2	95% CI	Length	$s^2 (\times 10^9)$	95% CI ($\times 10^9$)	Length ($\times 10^9$)
Chi-square	48.03	[46.22, 49.94]	3.72	16.89	[16.70, 17.08]	0.382
Normal	48.03	[45.34, 50.71]	5.36	16.89	[16.00, 17.78]	1.788
Pctl Boot	48.03	[45.44, 50.74]	5.30	16.89	[16.01, 17.81]	1.796
Stud Boot	48.03	[45.49, 50.90]	5.41	16.89	[16.04, 17.85]	1.811
Kurtosis $\hat{\kappa}$	2.15	—	—	41.77	—	—
Runtime	2.3s	—	—	258.5s	—	—

Note: s^2 is unbiased sample variance (SPDV). Length = upper – lower bounds.

Pctl Boot: Percentile bootstrap ($B = 5,000$); Stud Boot: Studentized SPDV bootstrap ($B = 5,000$ BMI / $B = 50,000$ CPS).

Data: BRFSS Louisiana 2022 ($n = 5,117$), CPS 2022 household income ($n = 60,000$ positive values).

[6]. Full code available at <https://github.com/asrivas18/SPDV-bootstrap> (Zenodo-archived upon acceptance).

5.12 Practical Implementation

The SPDV framework bridges theory and practice. The $\mathcal{O}(n)$ identity enables high-replication inference ($B = 50,000$) at scale ($n \approx 10^5$) on standard hardware, facilitating reproducible uncertainty quantification for policy applications.

6 Discussion and Concluding Remarks

This paper integrates classical U -statistic theory with modern computational practice for variance inference. While the pairwise representation of the sample variance is well known, its algorithmic implications for large-scale studentized bootstrap procedures have not been widely emphasized. By exploiting the algebraic equivalence between the SPDV representation and the standard variance formula, bootstrap variance inference can be implemented with linear-time complexity per resample, making high-replication studentized procedures feasible even for large datasets ($n \approx 10^6$, $B \geq 50,000$).

6.1 Principal Contributions

This work advances computational statistics for variance inference in three principal ways:

1. **Projection-based variance inference:** We formalize the sample variance as a non-degenerate second-order U -statistic through its pairwise representation (SPDV), yielding transparent asymptotics under minimal moment conditions ($\mathbb{E}[X^4] < \infty$; Theorem 2) and an explicit projection-based standard error.
2. **Scalable studentized bootstrap implementation:** By exploiting the classical $\text{SPDV} = s^2$ identity, we show how standard variance computations can be

used to implement bootstrap- t intervals in $O(Bn)$ time per resample, making high-replication ($B \geq 50,000$) studentized inference feasible even for very large samples ($n \approx 10^6$; Table 1).

3. **Applied guidance under skewness and heavy tails:** Through Monte Carlo experiments and two real-data examples, we document the finite-sample performance of chi-square, normal-approximation, percentile bootstrap, and studentized bootstrap intervals under skewed and heavy-tailed distributions, and provide practical recommendations for choosing B and for handling extreme tails in applied work.

6.2 Computational Implementation and Scalability

Direct evaluation of the pairwise variance representation, while pedagogically valuable, scales as $\mathcal{O}(n^2)$ and becomes prohibitive at modern data volumes. The identity-based approach exploits the exact algebraic equivalence to the standard variance formula, achieving $\mathcal{O}(n)$ per resample and providing multiple orders of magnitude speedup (Table 1). Studentized (bootstrap- t) intervals require standard error estimation per replicate. Double bootstrapping ($\mathcal{O}(B^2n)$ total) offers theoretical refinement but is impractical at scale. We recommend empirical fourth-moment estimation or jackknife-after-bootstrap as efficient $\mathcal{O}(Bn)$ alternatives.

Remark 6 (Practitioner Guidance). *The n -out-of- n bootstrap is asymptotically valid under $\mathbb{E}[X^4] < \infty$. For extreme heavy tails (empirical kurtosis $\hat{\kappa} > 20$, suspected power-law behavior), we suggest using the m -out-of- n bootstrap ($m \approx \sqrt{n}$) or subsampling, which retains $\mathcal{O}(m)$ efficiency via the identity.*

6.3 Limitations and Robustness

Methodological positioning and novelty. The pairwise representation of variance, its equivalence to the classical unbiased sample variance, the associated U-statistic structure, and the resulting asymptotic theory are all classical. Indeed, the asymptotic distribution of s^2 may be derived either through U-statistic theory or through the multivariate delta method. The contribution of the present work is therefore not a new distributional theory for the sample variance. Rather, the novelty lies in systematically integrating the classical SPDV identity with large-scale studentized bootstrap computation, explicit complexity analysis, and extensive empirical evaluation at modern data scales. By explicitly analyzing algorithmic complexity and demonstrating practical feasibility at $n \approx 10^6$ and $B \geq 50,000$ (Tables 1, we provide a computational bridge between classical variance theory and scalable distribution-free inference in contemporary settings.

Bootstrap consistency requires finite fourth moments. When $\mathbb{E}[X^4] = \infty$ (e.g., Pareto $\alpha \leq 4$), $\sqrt{n}$ -normality fails, risking significant undercoverage. In particular, the t_3 simulation results in Section 5 should be interpreted as empirical illustrations of finite-sample behavior under infinite fourth moments; they do not constitute a theoretical validation of the finite-fourth-moment bootstrap guarantees established in Theorem 2. Results for the t_3 design in our simulation study should therefore be interpreted as empirical evidence about finite-sample performance rather than as theoretical validation, since the finite-fourth-moment assumption underlying Theorem 2 does not hold for t_3 . Robust alternatives include:

- Absolute difference kernels ($|X_i - X_j|$; Gini-like measures).

- Rank-based dispersion (MAD, Q_n).
- Truncated pairwise moments to bound outlier influence.

These are particularly suited for heavy-tailed applications in wealth and financial data analysis. We emphasize that our empirical comparisons focus on classical variance estimators and bootstrap procedures, rather than on robust alternatives such as MAD-based or rank-based measures of dispersion. Incorporating SPDV-style scalable implementation into robust variance proxies (e.g., Gini-type kernels or truncated pairwise statistics) is an important direction for future work, particularly in settings with extreme tails where even finite-variance assumptions may fail.

The pairwise framework opens several promising directions for future work. A natural multivariate extension would replace the scalar variance kernel with the covariance kernel $h((x_1, y_1), (x_2, y_2)) = \frac{1}{2}(x_1 - x_2)(y_1 - y_2)$, enabling scalable studentized bootstrap inference for covariance matrices via the sample covariance identity. Generalization to higher-order U -statistics (e.g., for skewness or kurtosis) could follow similar linear-time identities, though identifying them remains challenging. Extensions to dependent data structures (time series, spatial, or clustered observations) via block or subsampling bootstrap would also benefit from the computational efficiency gains demonstrated here.

A detailed treatment of m -out-of- n bootstrap calibration and robust pairwise kernels in infinite-moment regimes is beyond the scope of this article and is left for future work.

6.4 Pedagogical Significance

The pairwise view recasts variance as the expected squared distance between two randomly drawn observations. This perspective aids instruction by illustrating computational trade-offs between quadratic and linear algorithms, failure modes of parametric assumptions under non-normality, and interpretability of variance in applied domains like public health (BMI) or economic dispersion.

A useful way to summarize the pairwise perspective is to view the variance as quantifying the average squared separation between observations. Equivalently, the quantity $\sqrt{2s^2}$ may be interpreted as a root-mean-square pairwise difference. In the BMI example, $\sqrt{2s^2} \approx 9.80 \text{ kg/m}^2$ represents the typical magnitude of separation between two randomly selected adults from the Louisiana sample. In the household income example, the variance summarizes the average squared difference between the incomes of two randomly chosen households, emphasizing dispersion in economic outcomes rather than deviations from a single reference value. Such interpretations help connect the variance to naturally meaningful pairwise comparisons and may aid communication of variability measures in applied settings, including public health, economics, and educational assessment.

6.5 Reproducibility and Software

Full R implementation (including all simulation experiments, real-data applications, data preprocessing scripts, and parallel bootstrap routines) is available at <https://github.com/asrivas18/SPDV-bootstrap>. The repository contains:

- Scripts to reproduce the Monte Carlo coverage results (Table 2 and Figure 2), using `foreach`, `doParallel`, and `doRNG` for reproducible parallel execution with controlled random seeds.

- Routines for the BMI and CPS examples in Section 5, together with preprocessed subsets of the public BRFSS and CPS data.
- Parallelized studentized bootstrap functions supporting multi-core execution.

The repository will be permanently archived via Zenodo upon acceptance (DOI forthcoming). The online supplement includes additional computational benchmarks, including detailed timing tables corresponding to the runtime summaries referenced in Sections 4 and 5. All code was developed and tested under R 4.4.1; parallel backends enable efficient reproduction of large-scale experiments on standard multi-core hardware.

6.6 Concluding remarks

The pairwise U -statistic framework unifies interpretive clarity, theoretical rigor, and computational scalability for variance estimation. By enabling scalable studentized bootstrapping for large modern datasets, it positions variance inference as a practical, distribution-free tool for modern data analysis in biostatistics, economics, and beyond.

Choice of bootstrap replications (B). In the simulation settings examined here (sample sizes $n = 20$ to 100 , various tail behaviors), stable coverage and interval width were achieved with B in the range $1,000$ to $5,000$ (see also Table 3 for sensitivity under heavy tails). For most practical applications with moderate computational resources, $B = 2,000$ to $5,000$ provides a good balance between Monte Carlo error and runtime. In large-scale settings where parallel hardware makes computation inexpensive (e.g., multi-core servers or clusters), increasing to $B = 10,000$ – $50,000$ further reduces Monte Carlo variability in the quantiles and improves second-order accuracy of studentized intervals, as demonstrated by the high-replication benchmarks in Tables 1.

Data Availability Statement

All simulation code, real-data preprocessing scripts, and preprocessed BRFSS subset are openly available at <https://github.com/asrivast18/SPDV-bootstrap> (Zenodo DOI to be assigned upon acceptance). The 2022 Louisiana BRFSS microdata are publicly available from the Louisiana Department of Health.

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Declaration of interest

The authors report there are no competing interests to declare.

A Technical Proofs and Details

Assumption A1 (Finite Fourth Moment) Throughout the asymptotic analysis, we assume $\mathbb{E}[(X - \mu)^4] < \infty$. This condition ensures non-degeneracy of the U -statistic and validity of $\sqrt{n}$ -asymptotic normality.

A.1 Proof of Theorem 1: SPDV Identity

Proof. Consider the sum of squared pairwise differences. By symmetry of unordered pairs, we have

$$\sum_{1 \leq i < j \leq n} (X_i - X_j)^2 = \frac{1}{2} \sum_{i=1}^n \sum_{j=1}^n (X_i - X_j)^2. \quad (2)$$

Expanding the square yields

$$\begin{aligned} \sum_{i=1}^n \sum_{j=1}^n (X_i^2 - 2X_i X_j + X_j^2) &= n \sum_{i=1}^n X_i^2 - 2 \left(\sum_{i=1}^n X_i \right) \left(\sum_{j=1}^n X_j \right) + n \sum_{j=1}^n X_j^2 \\ &= 2n \sum_{i=1}^n X_i^2 - 2n^2 \bar{X}^2 = 2n \sum_{i=1}^n (X_i - \bar{X})^2. \end{aligned} \quad (3)$$

It follows that

$$\sum_{1 \leq i < j \leq n} (X_i - X_j)^2 = n \sum_{i=1}^n (X_i - \bar{X})^2. \quad (4)$$

Substituting into the SPDV definition:

$$\text{SPDV} = \frac{1}{\binom{n}{2}} \sum_{i < j} \frac{(X_i - X_j)^2}{2} = \frac{1}{\frac{n(n-1)}{2}} \cdot \frac{n}{2} \sum_{i=1}^n (X_i - \bar{X})^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2 = s^2. \quad (5)$$

□

A.2 Proof of Theorem 2: Asymptotics and Bootstrap Consistency

Proof. (i) The equivalence $\text{SPDV} = s^2$ follows from Theorem 1 and its derivation in Appendix A.1.

(ii) *Asymptotic normality.* The Hoeffding decomposition is

$$\text{SPDV} - \sigma^2 = \frac{2}{n} \sum_{i=1}^n \psi(X_i) + R_n, \quad R_n = o_p(n^{-1/2}), \quad (6)$$

where $\psi(x) = \mathbb{E}[h(x, X_2) \mid X_1 = x] - \sigma^2 = \frac{1}{2}[(x - \mu)^2 - \sigma^2]$ is the first-order projection of the kernel $h(x_1, x_2) = \frac{(x_1 - x_2)^2}{2}$. Normalizing gives

$$\sqrt{n}(\text{SPDV} - \sigma^2) = \frac{2}{\sqrt{n}} \sum_{i=1}^n \psi(X_i) + \sqrt{n}R_n, \quad (7)$$

where $\sqrt{n}R_n \xrightarrow{p} 0$. Given $\mathbb{E}[X^4] < \infty$, we have $\text{Var}(\psi(X)) = \frac{1}{4}(\mu_4 - \sigma^4) = \frac{\sigma^4}{4}(\kappa - 1) < \infty$, where $\kappa = \mu_4/\sigma^4$ is the kurtosis. By the Lindeberg–Lévy central limit theorem:

$$\frac{2}{\sqrt{n}} \sum_{i=1}^n \psi(X_i) \xrightarrow{d} \mathcal{N}(0, 4 \text{Var}(\psi(X))) = \mathcal{N}(0, \mu_4 - \sigma^4). \quad (8)$$

(iii) *Bootstrap consistency.* By Arcones and Giné (1992), the nonparametric bootstrap for non-degenerate U -statistics is consistent when $\mathbb{E}[|h(X_1, X_2)|^2] < \infty$. Here,

$$\mathbb{E}[|h(X_1, X_2)|^2] = \frac{1}{4}\mathbb{E}[(X_1 - X_2)^4] \leq 2\mathbb{E}[X_1^4] < \infty$$

under the fourth-moment assumption. Thus the conditional bootstrap distribution of $\sqrt{n}(\text{SPDV}^* - \text{SPDV})$ converges in probability to $\mathcal{N}(0, \mu_4 - \sigma^4)$.

Therefore, under Assumption A1, the nonparametric bootstrap consistently approximates the sampling distribution of SPDV, enabling valid percentile and studentized confidence intervals as stated in Theorem 2. $\square$

Remark 7 (Infinite Fourth Moments). *When $\mathbb{E}[X^4] = \infty$ (e.g., t_ν with $\nu \leq 4$), $\sqrt{n}$ -normality fails, and the standard n -out-of- n bootstrap lacks consistency guarantees. Formal validity requires modified schemes such as m -out-of- n bootstrap.*

Remark 8 (Hoeffding Decomposition (Full Form)). *The complete decomposition is*

$$\text{SPDV} - \sigma^2 = \frac{2}{n} \sum_{i=1}^n \psi(X_i) + R_n, \quad R_n = o_p(n^{-1/2}),$$

where $\psi(X_i) = \frac{(X_i - \mu)^2 - \sigma^2}{2}$. Normalizing by $\sqrt{n}$ yields

$$\sqrt{n}(\text{SPDV} - \sigma^2) = \frac{2}{\sqrt{n}} \sum_{i=1}^n \psi(X_i) + \sqrt{n} R_n,$$

with $\sqrt{n} R_n \xrightarrow{p} 0$ under finite fourth moments. The central limit theorem applies to the leading term, yielding the asymptotic normality in Theorem 2.

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